

Tristan Stoltz

Founder & Lead Architect — Luminous Dynamics Computational Consciousness | Distributed Governance | AI Safety

Founder of Luminous Dynamics. Building working implementations of consciousness theories as computational systems. Creator of Symthaea, a 1.13M-line Rust cognitive architecture implementing IIT, GWT, FEP, and HOT as an integrated real-time system. Creator of Mycelix, a 133-zome decentralized governance framework where consciousness metrics gate civic participation. US Army veteran (2015–2021). US/South African dual citizen.

Contact

Johannesburg, South Africa | (315) 879-2332 | tristan.stoltz@luminousdynamics.org
LinkedIn: [linkedin.com/in/tristan-stoltz](https://www.linkedin.com/in/tristan-stoltz) | GitHub: github.com/Luminous-Dynamics

Core Competencies

- **Computational Consciousness:** Integrated Information Theory (Phi), Global Workspace Theory, Free Energy Principle, Higher-Order Thought, Active Inference, Substrate Independence
- **Cognitive Architecture:** Hyperdimensional Computing (16,384D), Liquid Time-Constant Networks, Continuous-time Neural ODEs, Neuromodulation (9 transmitters), 12-region Actor Brain
- **AI Safety & Ethics:** Epistemic gating (structural hallucination prevention), Moral Algebra (92.9% Hendrycks ETHICS), Moral Topology (persistent homology), Consciousness-gated action selection
- **Systems Engineering:** Rust (2.15M+ authored LOC), TypeScript, Python, NixOS, Holochain/DLT
- **Cryptography & Security:** Zero-Knowledge Proofs, DID/VCs, Post-Quantum hardening (ML-KEM/ML-DSA), Byzantine-resilient Federated Learning

Research & Engineering

Symthaea — Computational Consciousness Architecture (2025–Present)

Luminous Dynamics — Founder & Lead Architect

A cognitive architecture where consciousness-like properties — integration, prediction, self-modeling, and value alignment — are the computational substrate, not an afterthought.

- Built unified HDC-LTC neuron: 16,384-dimensional binary hypervectors evolving via closed-form Liquid Time-Constant dynamics with $O(1)$ temporal jumps — a novel contribution unifying memory and computation
- Implemented spectral MIP-based Phi (IIT) running at 50Hz in a continuous cognitive loop; validated against exhaustive partition search (Pearson $r=0.99$, Spearman $\rho=0.93$ across 62 network configurations, 35 topologies)
- Engineered full active inference loop (perception → prediction → surprise → action) with moral free energy on a 7-dimensional manifold
- Built Broca language pipeline with epistemic gating — architecture physically prevents generation beyond measured confidence (21K LOC, 229+ tests)

- Developed Moral Algebra: 4 independent ethical signals combined via category-adaptive weighted voting — 92.9% accuracy on Hendrycks ETHICS benchmark (4 categories, 2K test)
- Created Psych-Bench: 143-benchmark suite across 27 cognitive domains (1,283 tests), validated against published human baselines; composite $z=+1.32$, normative $z=+1.08$ (99 benchmarks with published baselines)
- Evaluated 14 Butlin et al. (2023) consciousness indicators with score-derived thresholds (12+ Present, mean quality 0.81); 8-row mechanistic ablation matrix
- Built substrate independence framework evaluating consciousness feasibility across 8 substrate types with honest epistemic confidence overlays
- Scale: ~1,130K lines Rust (897K code), 21,516 tests, 55 workspace members, 98 feature flags

Mycelix — Consciousness-Gated Decentralized Governance (2025-Present)

Luminous Dynamics — Founder & Lead Architect

A 16-cluster fractal governance system on Holochain where consciousness metrics from Symthaea gate progressive civic participation.

- Designed and implemented 133+ zones across 16 domain clusters
- Engineered consciousness gating: 4D profile (identity/reputation/community/engagement) → 5 participation tiers (Observer → Guardian) with progressive vote weights — Sybil-resistant without KYC
- Implemented post-quantum cryptographic hardening (ML-KEM-768, ML-DSA-65) across all clusters
- Developed Byzantine-resilient Federated Learning protocol achieving 34% validated Byzantine tolerance
- Built Rust SDK (18 modules, 1,036+ tests) and TypeScript SDK (37 modules, 6,316 tests)
- Scale: ~785K lines Rust (643K code), 8,600+ workspace tests, open source (AGPL-3.0)

Publications & Research Outputs (In Preparation)

- “Symthaea: A Unified HDC-LTC-IIT Architecture for Computational Consciousness” — targeting PLoS Computational Biology
- “Psych-Bench: A 143-Benchmark Psychometric Suite for Evaluating Cognitive Architectures” — targeting Behavior Research Methods (40pp, arXiv preprint prepared)
- 18 Invention Disclosure Documents (203 claims across 3 tiers)

Prior Experience

Technical Account Manager — Centre Technologies (2022-2023)

- Led enterprise cloud migration projects (AWS/Azure), optimizing performance and reducing costs
- Engineered cybersecurity protocols and disaster recovery plans

System Administrator — US Army, Fort Drum, NY (2017-2021)

- Built PowerShell automation system achieving 80% improvement in operational efficiency

- Maintained server and workstation networks for mission-critical military operations

IT Support Team Lead — US Army, Fort Drum, NY (2016-2017)

- Led team of 9 technicians in high-pressure military support environment

Communication Transmissions Specialist — US Army, Fort Drum, NY (2015-2016)

- Managed SATCOM and MLOS communication systems ensuring secure operational connectivity

Military Training

- US Army Signal School — Communication Transmissions Specialist (25S), Fort Gordon, GA (2015)
- NCO leadership courses (2016-2020)

Independent Research Background

17 years (2009-present) of self-directed study across computational neuroscience, consciousness studies, and AI systems engineering. Primary literature: Tononi (IIT), Friston (FEP), Baars/Dehaene (GWT), Hasani (LTC/CfC), Kanerva (HDC), Butlin et al. (consciousness indicators), Ostrom (commons governance). Work validated through 42,000+ passing tests, benchmark reproduction against published baselines, and open-source peer review.